Report Bayesian Networks CPS

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Abstract

AMA-CPS Bayesian Network exam assignment: mini-project Individual work on the topics addressed in the Bayesian Network (BN) part of the course. The project will contribute to the final score through a weighted average. Please follow the instructions below. 1. Choose a CPS of your choice and a suitable sub-system to be modelled. 2. Define the dependability attributes of interest and related quantitative requirements. 3. Assume credible threat metrics (e.g., MTBF) for the basic components, and other possible external events. 4. If you start from Fault Tree model, justify the need for adding any common mode of failure, and/or any noisy-gates. Define the CPTs accordingly. 5. Build the BN model using a suitable tool such as Netica. It is recommended that the model should not exceed the size of 15 nodes. 6. Analyze the BN model using the techniques shown during the lectures. 7. Report and discuss the results. You are expected to write a brief report on those points (suggested length between 5 and 10 pages; no special requirements about the format or template), and a short presentation between 4 and 6 slides. Please submit the source files of your BN model as well as the report and the presentation (in pdf format). Submission and deadlines Information on how and when to submit the materials will be provided on the specific section of the course website. In case of any issues during submission, please email the docent. You must be prepared to present your slides during an oral exam day scheduled after mini-project submission deadline, if required.

1 Introduction

The goal of the proposed Bayesian Network (BN) is to estimate the probability of *Loss of Controlled Flight* (LCF) for an Unmanned Aerial System (UAS). The node LCF represents the top event of the system, corresponding to a catastrophic loss of control that may lead to mission failure.

The structure of the BN has been derived from the logical decomposition of the Fault Tree (FT) developed in the reference article [1], and it incorporates a set of basic events representing software faults, environmental hazards, human errors, and communication issues. Probabilistic parameters for each basic event have been obtained from the literature 1. The nominal values were used as prior probabilities in the BN.

The Bayesian Network formalism enables:

- coherent representation of uncertainty and dependencies among subsystems;
- *a priori* evaluation of the probability of LCF;
- *a posteriori* analysis to identify the most probable causes of LCF;
- sensitivity analysis and Most Probable Explanation (MPE);
- derivation of reliability metrics such as MTBF and $R(t)$.

Basic Events	P_{min}	P_{max}	P_{mean}
No update from other drone's data	5.5E-02	3.7E-01	1.4E-01
Snipping from enemy	6.0E-03	1.8E-02	8.0E-03
Electromagnetic pulse	4.6E-03	2.6E-02	1.8E-02
Bad maintenance	4.0E-03	2.2E-02	1.2E-02
Software not updated	4.0E-03	2.0E-02	6.0E-03
OS problem	2.0E-03	8.0E-03	3.0E-03
Virus/Malware	6.0E-03	1.4E-02	1.0E-02
Human Fatigue	1.0E-02	4.0E-02	2.0E-02
Human Inexperience	1.6E-02	6.0E-02	2.4E-02
Misjudgment of weather	1.1E-02	3.6E-02	2.2E-02
Obstacles	1.5E-01	4.0E-01	3.5E-01
Decoding error	1.2E-03	6.8E-03	4.6E-03
Attackers	2.4E-02	8.0E-02	3.0E-02
Bad weather	1.4E-02	6.8E-02	4.7E-02
Atmospheric attenuation	8.0E-03	2.6E-02	1.2E-02
Large distance from transmitter	2.9E-03	1.1E-01	4.3E-03

Figure 1: Probabilities of basic events derived from the article [1].

2 System Modelling

2.1 Basic Events

The BN includes a set of root nodes representing elementary failures or contextual conditions. Each basic event is modelled as a binary discrete node with states `True` and `False`. The following basic events were considered:

- **VirusMalware**
- **OSProblem**
- **SoftwareNotUpdated**
- **DecodingError**

- **LargeDistanceFromTransmitter**
- **AtmosphericAttenuation**
- **Obstacles**
- **BadWeather**
- **HumanInexperience**
- **BadMaintenance**

Each prior probability was set using the nominal value *Pnom* extracted from the reference dataset. These values represent the likelihood of each basic event occurring during a standard mission.

2.2 Intermediate Nodes and Top Event

To reflect the hierarchical structure of the FT and to aggregate related basic events, four intermediate nodes and the top event were introduced:

- **Software Failure:** represents failures on the onboard software or operating system. Parents: OS Problem, Virus Malware and Software Not Updated.
- **Loss Of Communication:** represents the loss of the communication link between the drone and the ground station.
- **Environmental:** represents hazardous environmental conditions affecting flight safety. Parents: Bad Weather, Obstacles and Atmospheric Attenuation.
- **GSC Operational:** represents operational errors at the Ground Station Control (GSC). Parents: Human Inexperience and Bad Maintenance.
- **Loss of Controlled Flight (LCF):** representing the loss of control of the aircraft. Parents: Software Failure, Loss Of Communication, Environmental, GSC Operational.

The states of every node are: `True` and `False` and every CPT is modelled using Noisy-Or function.

3 Assumptions and Modelling Choices

The construction of the Bayesian Network relies on a set of modelling assumptions that define the interpretation of probabilities, the temporal scope of the analysis, and the dependency structure between the nodes. These assumptions are consistent with the reference Fault Tree and with the reliability-oriented modelling approach adopted in the article [1].

3.1 Mission Time and Interpretation of Probabilities

All prior probabilities assigned to basic events represent the likelihood of the corresponding failure occurring during a *single mission*. The mission is assumed to follow a standard operational profile, with fixed duration and environmental exposure.

The BN does not explicitly model the temporal evolution of failures; instead, it provides a static probabilistic assessment for a mission. Reliability metrics such as Mean Time Between Failure (MTBF) are derived by assuming an exponential failure distribution:

$$P(t) = 1 - e^{-\lambda t},$$

where t is the mission duration.

3.2 Independence Assumptions

Basic events are assumed to be mutually independent unless explicitly connected through intermediate nodes. This assumption is standard in Bayesian reliability modelling and reflects the structure of the underlying Fault Tree. Dependencies among subsystems (e.g., environmental conditions affecting communication) are captured through intermediate nodes rather than direct links between basic events.

3.3 Use of Noisy-OR Models

Intermediate nodes and the top event LCF are modelled using Noisy-OR conditional probability tables. This choice is motivated by:

- the need to represent causal contributions of multiple parent failures without enumerating all combinations,
- the interpretability of causal strengths associated with each parent,
- the consistency with the logical structure of the Fault Tree.

The Noisy-OR model assumes that each parent can independently trigger the child node, with a certain causal strength Pi, while the leak probability accounts for background causes that might trigger the event even when all modeled parents are absent. This provides a compact and realistic representation of failure propagation in uncertain environments.

3.4 Top Event Interpretation

Since the LCF corresponds to a non-recoverable failure, the model does not include repair processes or recovery times. As a consequence:

- MTTR is not meaningful in this context,
- Availability is not considered as a primary metric,
- Reliability $R(t)$ and MTBF are the relevant performance indicators.

3.5 Scope and Limitations

The model focuses on the dominant contributors to LCF: software faults, environmental hazards, communication issues, and human errors. Other factors (e.g., hardware degradation, battery failures, aerodynamic anomalies) are not included to maintain tractability and to remain consistent with the reference Fault Tree. The BN provides a probabilistic assessment rather than a deterministic diagnosis, and results depend on the accuracy of the prior probabilities extracted from literature.

4 Probabilistic Parametrization of Basic Events

Each basic event is modelled as a binary node with states `True` and `False`, where:

$$P(\texttt{True}) = P_{\text{mean}}, \qquad P(\texttt{False}) = 1 - P_{\text{mean}}.$$

Table 1 reports the probabilistic parameters used for each basic event.

This parametrization ensures consistency with the reference Fault Tree and provides a realistic representation of the uncertainty associated with UAS operations.

Table 1: Probabilistic parameters of basic events.

Basic Event	P_{nom}
Bad maintenance	2.2×10^{-2}
Software not updated	2.0×10^{-2}
OS problem	8.0×10^{-3}
Virus/Malware	1.4×10^{-2}
Human inexperience	6.0×10^{-2}
Obstacles	4.0×10^{-1}
Decoding error	6.8×10^{-3}
Bad weather	6.8×10^{-2}
Atmospheric attenuation	2.6×10^{-2}
Large distance from transmitter	1.1×10^{-1}

5 A Priori Analysis

The *a priori* analysis consists in evaluating the probability of the top event LCF before introducing any evidence in the Bayesian Network. This corresponds to computing the unconditional probability of loss of controlled flight based solely on the prior probabilities of the basic events and on the causal structure encoded in the intermediate nodes.

Using the nominal probabilities P_{nom} for all basic events, the BN produces the following *a priori* estimate:

$$P(\text{LCF} = \texttt{True}) \approx 0.329,$$

which indicates that, under nominal operating conditions, the probability of losing control during a standard mission is approximately 32.9%. The complementary probability,

$$P(\text{LCF} = \texttt{False}) \approx 0.671,$$

represents the likelihood of completing the mission without experiencing a catastrophic loss of control.

This value reflects the combined influence of the intermediate nodes *SoftwareFailure*, *LossOfCommunication*, *Environmental*, and *GSCOperational*, each of which aggregates multiple basic events through Noisy-OR conditional probability models. In particular, the relatively high nominal probabilities associated with environmental hazards (e.g., *Obstacles*, *BadWeather*, *AtmosphericAttenuation*) significantly contribute to the overall risk.

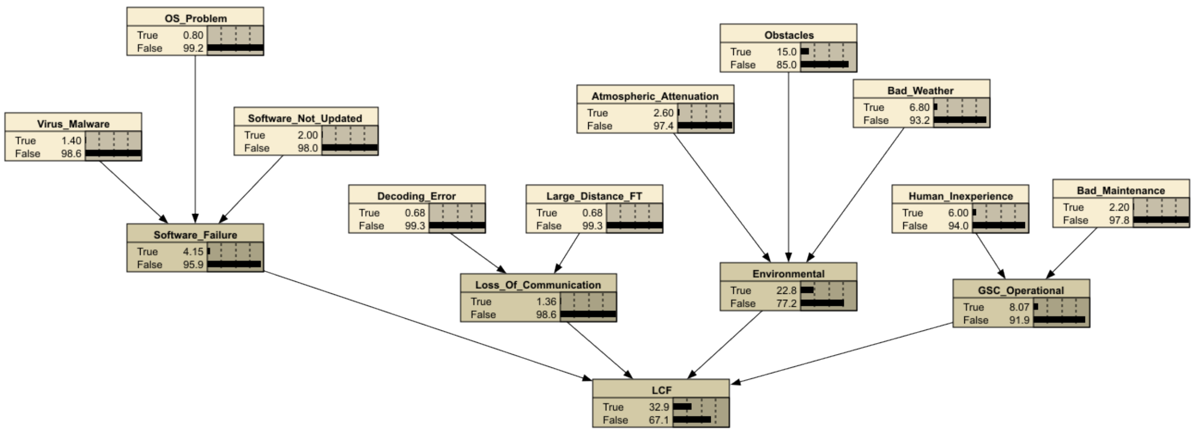

Figure 2: A priori analysis results with mean probabilities

5.1 Interpretation of the A Priori Result

The *a priori* probability of LCF should be interpreted as the expected mission-level failure probability in the absence of any additional information about the system state. This value serves as a baseline for subsequent analyses:

- **A posteriori analysis**, where evidence on LCF is introduced to identify the most probable causes of failure;
- **Sensitivity analysis**, which quantifies the influence of each node on the uncertainty of LCF;

The *a priori* estimate also provides the basis for deriving reliability metrics such as the failure rate λ and the Mean Time Between Failures (MTBF), assuming an exponential failure distribution.

6 A Posteriori Analysis

The *a posteriori* analysis consists in conditioning the Bayesian Network on the occurrence of the top event, i.e., setting:

$$\mathrm{LCF} = \texttt{True},$$

and computing the posterior probabilities of all intermediate and basic events. This analysis identifies the most likely causes of the loss of controlled flight and provides a probabilistic ranking of contributing factors.

Using the nominal prior probabilities and the Noisy-OR structure of the BN, the posterior probabilities of the intermediate nodes are reported in Table 2.

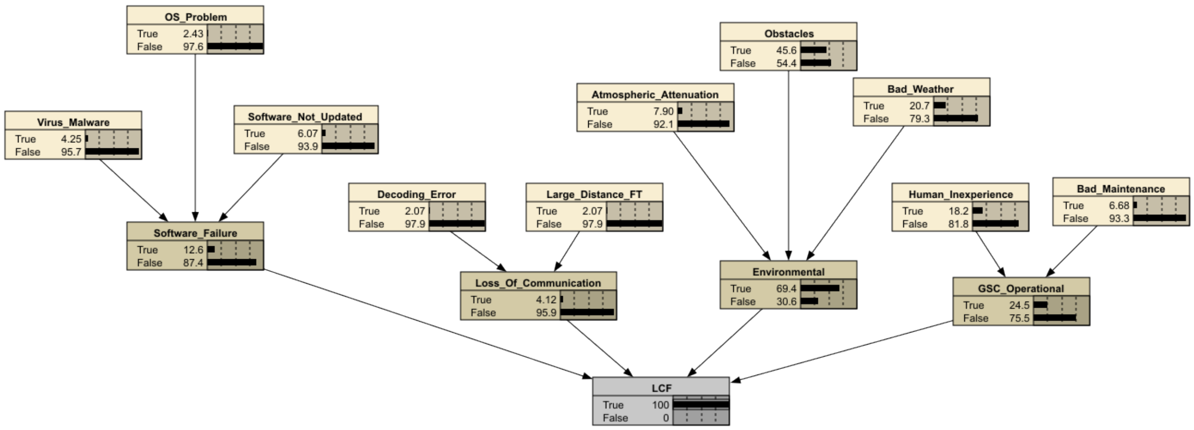

Figure 3: A posteriori analysis results with nominal probabilities

Table 2: Posterior probabilities of intermediate nodes given LCF = True.

Node	$P(\cdot \mid \text{LCF} = \texttt{True})$
Environmental (Hazardous)	0.694
GSCOperational (Error)	0.245
SoftwareFailure (Failure)	0.126
LossOfCommunication (Lost)	0.0412

The results show that **environmental hazards** represent the dominant contributor to LCF, with a posterior probability of approximately 69.4%. This is consistent with the relatively high prior probabilities associated with *Obstacles*, *BadWeather*, and *AtmosphericAttenuation*. Operational errors at the Ground Station Control also play a significant role (24.5%), primarily driven by *HumanInexperience*. Software-related failures contribute to a lesser extent (12.6%), while loss of communication has a marginal impact (4.12%).

6.1 Interpretation of Posterior Results

The posterior distribution highlights the most probable causal pathways leading to LCF. In particular:

- **Environmental conditions** are the most influential factor, suggesting that adverse weather and physical obstacles are the primary drivers of mission failure.

- **Human-related errors** at the GSC represent the second most relevant contributor, reflecting the importance of operator expertise.
- **Software failures** have a non-negligible but secondary impact.
- **Communication failures** are unlikely to be the root cause of LCF under nominal conditions.

These findings are consistent with the results obtained in the reference thesis and confirm that the BN correctly captures the dominant failure mechanisms affecting UAS operations.

The *a posteriori* analysis also provides the foundation for further diagnostic evaluations, such as the Most Probable Explanation (MPE), which identifies the single most likely configuration of system states leading to LCF.

7 Sensitivity to Findings

The *Sensitivity to Findings* analysis quantifies how much each node in the Bayesian Network contributes to reducing the uncertainty of the top event LCF. For each node X, Netica computes the mutual information $I(\mathrm{LCF}; X)$ and the percentage of belief variance explained by X. This analysis identifies which variables are the most informative for predicting the occurrence of LCF.

Table 3 reports the sensitivity values obtained by selecting LCF as the target node.

Table 3: Sensitivity of LCF to findings on other nodes.

Node	**Mutual Info**	**Percent**	**Variance of Beliefs**
LCF	0.91418	100	0.2208508
Environmental	0.48251	52.8	0.1331663
Obstacles	0.28245	30.9	0.0793907
GSC_Operational	0.14007	15.3	0.0394817
Bad_Weather	0.11648	12.7	0.0328239
Human_Inexperience	0.10192	11.1	0.0287158
Software_Failure	0.06911	7.56	0.0194540
Atmospheric_Attenuation	0.04270	4.67	0.0120091
Bad_Maintenance	0.03599	3.94	0.0101200
Software_Not_Updated	0.03266	3.57	0.0091812
Virus_Malware	0.02273	2.49	0.0063878
Loss_Of_Communication	0.02200	2.41	0.0061814
OS_Problem	0.01292	1.41	0.0036281
Decoding_Error	0.01097	1.20	0.0030801
Large_Distance_FT	0.01097	1.20	0.0030801

7.1 Interpretation

The results clearly indicate that the most informative variables for predicting LCF are those related to environmental conditions. In particular:

- **Environmental** is the dominant contributor, explaining approximately 52.8% of the belief variance of LCF.
- The basic event **Obstacles** is the single most informative leaf node, with 30.9%.
- **BadWeather** and **AtmosphericAttenuation** also provide significant information, confirming the strong influence of environmental hazards.

- **GSCOperational** and **HumanInexperience** contribute moderately, reflecting the relevance of human factors.
- **Software failures** and communication issues exhibit lower sensitivity values, indicating a limited impact on the uncertainty of LCF under nominal conditions.

7.2 Discussion

The sensitivity analysis confirms that the BN is strongly driven by environmental factors, which dominate both the *a priori* and *a posteriori* behaviour of the model. This is consistent with the structure of the reference Fault Tree and with the results obtained in the thesis, where environmental hazards were identified as the primary contributors to the loss of controlled flight.

Moreover, the analysis highlights which variables would be most valuable to monitor in real-time to reduce uncertainty on LCF. In particular, sensors providing information on obstacles, weather conditions, and signal attenuation would yield the greatest reduction in predictive uncertainty.

8 Most Probable Explanation (MPE)

The *Most Probable Explanation* (MPE) analysis identifies the single most likely configuration of all nodes in the Bayesian Network that leads to the observation of the top event. While the *a posteriori* analysis provides marginal probabilities for each node conditioned on LCF, the MPE returns a complete joint assignment of states that maximizes:

$$P(\mathbf{X} \mid \mathrm{LCF} = \texttt{True}),$$

where $\mathbf{X}$ denotes the set of all nodes in the network.

This analysis provides a concrete scenario representing the most plausible chain of events that results in the loss of controlled flight.

8.1 MPE Results

By conditioning the BN on LCF = `True` and computing the MPE, the most probable explanation for the occurrence of LCF is characterized by the following configuration of nodes:

- **Obstacles = True** with posterior probability ≈ 1.00,
- **Environmental = True** with posterior probability ≈ 1.00,

indicating that the presence of obstacles and hazardous environmental conditions is almost certainly part of the most probable failure scenario.

Other nodes show a non-negligible increase in their posterior probability of being in the failed state when conditioning on LCF:

- **Bad_Weather = True** with posterior probability ≈ 0.413,
- **Human_Inexperience = True** with posterior probability ≈ 0.362,
- **GSC_Operational = True** with posterior probability ≈ 0.362,
- **Atmospheric_Attenuation = True** with posterior probability ≈ 0.151,
- **Bad_Maintenance = True** with posterior probability ≈ 0.127,
- **Software_Failure = True** with posterior probability ≈ 0.116,

- **Virus_Malware = True** with posterior probability ≈ 0.0805,
- **OS_Problem = True** with posterior probability ≈ 0.0457,
- **Software_Not_Updated = True** with posterior probability ≈ 0.116,
- **Loss_Of_Communication = True** and **Large_Distance_FT = True** with posterior probability ≈ 0.0388,
- **Decoding_Error = True** with posterior probability ≈ 0.0388.

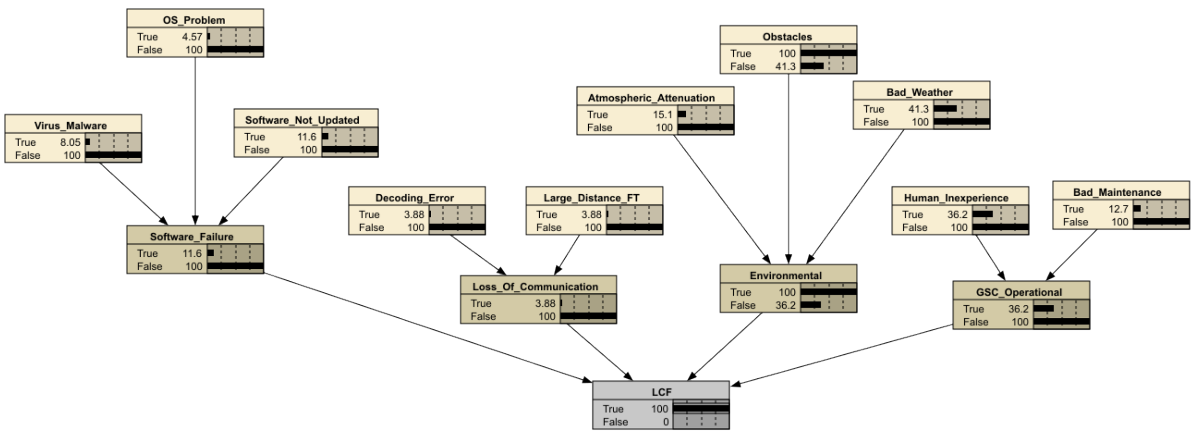

Figure 4: Most probable exaplanation analysis results.

Although many of these probabilities remain below 0.5 and therefore their most likely state in the strict MPE configuration is still the non-failed one, their posterior increase with respect to the prior clearly indicates that they are plausible contributors to the observed failure.

8.2 Interpretation

The MPE analysis, together with the posterior distributions under LCF = `True`, highlights a failure scenario strongly dominated by environmental factors:

- **Obstacles** and the aggregated node **Environmental** are almost certainly active in the failure scenario, confirming that environmental hazards are the primary drivers of LCF.
- **Bad_Weather**, **Atmospheric_Attenuation** and **Obstacles** jointly contribute to the activation of the Environmental node.

- **Human_Inexperience** and **GSC_Operational** show significantly increased posterior probabilities, indicating that operator-related errors are also likely to play a role in the loss of controlled flight.
- Software and communication-related failures exhibit smaller, but still non-negligible, posterior probabilities, suggesting that they may act as secondary contributors rather than primary root causes.

8.3 Discussion

The MPE conditioned on LCF = `True` provides a coherent and interpretable causal scenario that is fully consistent with the results of the *a posteriori* and sensitivity analyses. Environmental hazards, in particular obstacles, emerge as the dominant contributors to the loss of controlled flight, with human operational factors playing an important supporting role. This confirms that improvements in environmental sensing, obstacle detection, and operator training are key levers for reducing the likelihood of catastrophic mission failure.

This analysis reinforces the conclusion that improving environmental sensing, obstacle detection, and operator training would yield the greatest reduction in the likelihood of catastrophic mission failure.

9 MTBF and Reliability

The Bayesian Network provides an *a priori* estimate of the probability of Loss of Controlled Flight (LCF) during a standard mission. Using the nominal probabilities of the basic events, the unconditional probability of failure is:

$$P(\text{LCF} = \texttt{True}) \approx 0.329.$$

To derive reliability metrics, this probability is interpreted as the likelihood of experiencing a catastrophic failure within a single mission of duration T. Assuming that failures follow an exponential distribution, the mission-level failure probability is related to the failure rate λ by:

$$P(\text{failure in } T) = 1 - e^{-\lambda T}.$$

By setting $T = 1$ mission and solving for λ, we obtain:

$$0.329 = 1 - e^{-\lambda} \quad \Rightarrow \quad \lambda = -\ln(0.671) \approx 0.399 \text{ failures/mission}.$$

9.1 Mean Time Between Failures (MTBF)

The Mean Time Between Failures is defined as the inverse of the failure rate:

$$\text{MTBF} = \frac{1}{\lambda}.$$

Using the estimated value of λ:

$$\text{MTBF} \approx \frac{1}{0.399} \approx 2.5 \text{ missions}.$$

This value represents the expected number of missions that the UAS can complete before experiencing a loss of controlled flight under nominal operating conditions.

9.2 Reliability Function

The reliability of the system over time is given by the exponential reliability function:

$$R(t) = e^{-t/\mathrm{MTBF}},$$

where t is expressed in mission units. Using the computed MTBF, the reliability for different mission horizons is:

$$R(1) = e^{-1/2.5} \approx 0.670,$$

$$R(5) = e^{-5/2.5} \approx 0.135,$$

$$R(10) = e^{-10/2.5} \approx 0.018.$$

These values indicate that the probability of completing multiple consecutive missions without experiencing LCF decreases rapidly, reflecting the relatively high mission-level failure probability.

9.3 Discussion

The MTBF and reliability results must be interpreted in the context of the Bayesian Network model. Since LCF represents a catastrophic and non-recoverable event, the model does not include repair processes or MTTR considerations. Therefore:

- MTTR is not meaningful for this system,
- Availability is not a relevant metric,
- Reliability and MTBF provide the appropriate performance indicators.

The derived MTBF of approximately 2.5 missions is consistent with the high posterior influence of environmental hazards and operator errors, as highlighted in the sensitivity and MPE analyses. These results reinforce the need for improved environmental sensing, obstacle avoidance, and operator training to increase mission reliability.

10 Conclusions

This work presented a Bayesian Network model for estimating the probability of *Loss of Controlled Flight* (LCF) in an Unmanned Aerial System (UAS). Starting from the structure of the reference Fault Tree, the BN integrates software failures, environmental hazards, communication issues, and human operational errors into a unified probabilistic framework capable of capturing uncertainty and causal dependencies among subsystems.

The *a priori* analysis yielded an unconditional probability of $P(\mathrm{LCF}) \approx 0.329$ under nominal operating conditions, highlighting the significant impact of environmental exposure and operator-related factors on mission safety. The *a posteriori* analysis confirmed that environmental hazards and GSC operational errors are the dominant contributors to LCF, while software and communication failures play a secondary role.

The *Sensitivity to Findings* analysis further reinforced these results, showing that the nodes *Environmental*, *Obstacles*, and *BadWeather* are the most informative variables for predicting LCF. Human-related factors also contribute meaningfully, whereas software and communication events exhibit lower sensitivity values.

The *Most Probable Explanation* (MPE) identified a coherent and realistic failure scenario dominated by adverse environmental conditions and operator inexperience, consistent with both the sensitivity analysis and the structure of the underlying Fault Tree.

Finally, reliability metrics were derived by interpreting the mission-level failure probability within an exponential failure model. The resulting failure rate $\lambda \approx 0.399$ failures/mission corresponds to an MTBF of approximately 2.5 missions, indicating that the probability of completing multiple consecutive missions without LCF decreases rapidly.

Overall, the Bayesian Network provides a transparent and analytically robust framework for assessing UAS mission risk. Future extensions may include the integration of hardware degradation models, dynamic Bayesian Networks for temporal reasoning, or real-time evidence updates during mission execution.

References

[1] R. Abdallah, R. Kouta, C. Sarraf, J. Gaber, and M. Wack, "Fault tree analysis for the communication of a fleet formation flight of uavs," in 2017 2nd international Conference on system Reliability and safety (ICSRS), IEEE, 2017, pp. 202–206.